The Lemma of Gabber

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The principal result of this exposé, Theorem 1.4, is due to O. Gabber.

1 Statements

1.1 In this exposé we systematically work with algebraic spaces, rather than only with schemes. This raises a problem of references. We shall allow ourselves to refer to an article or theorem proved for schemes and apply it to algebraic spaces whenever the same proof also proves the assertion for algebraic spaces. A proof becomes problematic when it uses the fact that every point has an affine neighbourhood and when "neighbourhood" cannot be replaced by "étale neighbourhood".

This leads us to speak of an *abelian algebraic space* A over S, where one would ordinarily speak of an abelian scheme over S: this means a commutative S-algebraic space in groups

$$f : A \longrightarrow S$$

with f proper, flat, and of finite presentation, and with geometric fibres abelian varieties. We shall call A a *semi-abelian algebraic space* over S if it is a commutative S-algebraic space in groups $f : A \to S$, with f separated, flat, of finite presentation, and with geometric fibres extensions of abelian varieties by tori.

Note, however, the following points.

(a) If S is the spectrum of a field, every algebraic space in groups over S is a scheme, and indeed is quasi-projective.

(b) If S is a trait, every algebraic space in groups G over S, locally of finite type and separated over S, is a scheme. This is proved for the smooth case in Raynaud's work on Néron models.

(c) If S is normal integral, with generic point η, and if A is an abelian algebraic space over S, Grothendieck's arguments show that every polarization of A_η extends uniquely to a polarization of A over S. It follows that A is projective over S, hence in particular that A is a scheme if S is a scheme.

An algebraic space $f : X \to S$, flat of finite presentation, separated over S, and with proper and connected geometric fibres, is proper over S. By localization on S and passage to limits, one reduces to the case where S is the spectrum of a noetherian local ring; by faithfully flat descent one may suppose the ring complete, and then one applies EGA III, first part, 5.5. In particular, a semi-abelian algebraic space over S all of whose fibres are abelian varieties is abelian over S.

Unless explicitly stated otherwise, all schemes and algebraic spaces considered below are quasi-compact and quasi-separated.

1.2 Let U be a dense open subspace of a normal algebraic space S, and let A be an abelian algebraic space over U. A *semi-abelian prolongation* of A over S is a semi-abelian algebraic space P over S together with an isomorphism

$$A \xrightarrow{\sim} P|_U.$$

Theorem 1.1 (Raynaud)**.** *Under the preceding hypotheses, if A admits a semi-abelian prolongation, then that prolongation is unique up to unique isomorphism.*

Localization and passage to limits reduce the proof to the case where S is a noetherian scheme. If S is a trait and U is its generic point, every semi-abelian prolongation P of A coincides with the identity component N^0 of the Néron model N of A: since P is smooth with connected fibres, there is a morphism $P \to N^0$ inducing the identity on the generic fibre, and SGA 7 IX, 3.1 shows that it is an isomorphism. For normal noetherian S, two prolongations coincide on an open $V \supset U$ whose complement has codimension at least two. Raynaud's extension theorem then gives the result. A correction to this argument is supplied in the note at the end of the exposé.

1.3

Remark 1.2. *If P and Q are two semi-abelian algebraic spaces over S, with S normal and integral and with generic fibres P_η and Q_η abelian varieties over $k(\eta)$, the same argument gives*

$$\mathrm{Hom}_S(P, Q) \hookrightarrow \mathrm{Hom}_\eta(P_\eta, Q_\eta).$$

Remark 1.3. *The normality hypothesis on S is indispensable. Let S be a nodal cubic over an algebraically closed field, obtained from $\mathbf{P}^1$ by identifying 0 and ∞ in one point p; let $U = S - \{p\}$; and let A be the constant elliptic curve E over U. To give a prolongation of A is to give a prolongation over $\mathbf{P}^1$ and a gluing isomorphism between the fibres at 0 and at ∞. The prolongation over $\mathbf{P}^1$ is unique, namely $E \times \mathbf{P}^1$, but the gluing isomorphism can be chosen in at least two ways because $\mathrm{Aut}(E)$ contains $\{\pm 1\}$.*

The theorem should nevertheless remain true when U is schematically dense in S and S is only assumed geometrically unibranch. For general S, two prolongations inducing the same prolongations after the relevant normalizations should be isomorphic.

1.4 To avoid references that are not available, and because this case suffices for the intended application, we state Gabber's lemma under hypotheses that are probably unnecessarily restrictive.

Theorem 1.4 (Gabber's lemma)**.** *Let U be a dense open subscheme of an excellent normal integral scheme S, and let A be an abelian algebraic space over U. There exists a proper surjective morphism*

$$f : S' \longrightarrow S$$

such that the inverse image of A over $f^{-1}(U)$ extends to a semi-abelian algebraic space over S'.

1.5 If there existed a moduli space for semi-abelian algebraic spaces, the semi-stable reduction theorem together with the valuative criterion of properness would show that this moduli object is proper over $\mathrm{Spec}(\mathbf{Z})$, and Gabber's lemma would follow immediately. But we are far from such a situation: the objects considered have automorphisms, sometimes infinitely many, and worse, a semi-abelian algebraic space over a complete discrete valuation ring is not determined by the associated formal algebraic space. If the special fibre is isomorphic to $\mathbf{G}_m$, the reduction modulo $\mathfrak{m}^k$ is isomorphic to that of $\mathbf{G}_m$ for every k by rigidity of tori, without A necessarily being a torus.

The bad behaviour of semi-abelian spaces with respect to formal completion comes from the fact that they are not necessarily proper. For curves the situation is better because one has the notion of a stable curve of genus g. Such curves are proper and, for $g \geq 2$, have only finitely many automorphisms. Because of finite automorphism groups, stable curves of genus $g \geq 2$ do not give a fine moduli space, but they do give a proper algebraic stack over $\mathrm{Spec}(\mathbf{Z})$ by the valuative criterion of properness and the stable reduction theorem for curves.

1.6

Lemma 1.5. *Let U be a quasi-compact dense open subspace of an algebraic space S, and let $X \to S$ be a family of stable curves of genus $g \geq 2$ parametrized by U. There exists a proper surjective morphism*

$$f : S' \longrightarrow S$$

such that the inverse image of X over $f^{-1}(U)$ extends to a family of stable curves over S'.

Proof. Fix $g \geq 2$, and for brevity say simply "stable curve" for stable curve of genus g. Let $\mathcal{M}$ be the algebraic stack classifying stable curves. It is proper over $\operatorname{Spec}(\mathbf{Z})$. There is therefore a proper scheme S_0 over $\operatorname{Spec}(\mathbf{Z})$ and a surjective morphism $S_0 \to \mathcal{M}$; equivalently, S_0 is endowed with a family of stable curves $u_0 : X_0 \to S_0$ such that every stable curve over an algebraically closed field occurs as a fibre of X_0/S_0.

Over $U \times S_0$ we have the two stable curves $\operatorname{pr}_1^* X$ and $\operatorname{pr}_2^* X_0$. Let

$$I = \operatorname{Isom}(\operatorname{pr}_1^* X, \operatorname{pr}_2^* X_0).$$

This algebraic space is finite over $U \times S_0$ and maps onto U. Let Γ be its schematic image in $U \times S_0$, and let $\overline{I}$ and $\overline{\Gamma}$ be the schematic closures inside $S \times S_0$:

$$\begin{array}{ccccc} I & \longrightarrow & \Gamma & \hookrightarrow & U \times S_0 \\ \downarrow & & \downarrow & & \downarrow \\ \overline{I} & \longrightarrow & \overline{\Gamma} & \hookrightarrow & S \times S_0. \end{array}$$

The algebraic space $\overline{I}$, endowed with the proper morphism

$$\overline{I} \longrightarrow \overline{\Gamma} \hookrightarrow S \times S_0 \longrightarrow S,$$

is the desired S'. The inverse image over $\overline{I}$ of the universal family X_0/S_0 extends the inverse image over I of X/U. $\square$

In Section 3 we shall demystify the use of algebraic stacks in this proof.

1.7 The principle of the proof of Gabber's lemma is to reduce to the case of curves by using Jacobians. We record some preliminary remarks.

(a) As explained in 1.1(a), A/U is automatically an abelian scheme.

(b) Given $U \hookrightarrow S$, A/U, and $f : S' \to S$ as in Theorem 1.4, every proper surjective $f_1 : S'' \to S$ lying over S' again satisfies the conclusion. Applying Chow's lemma and normalizing, if the theorem holds at all then it holds with $f : S' \to S$ projective and surjective, S' integral and normal.

(c) It suffices to verify the theorem locally on S. If S is covered by finitely many open subspaces S_i, and over each S_i one has a solution $S'_i \to S_i$, one can find a projective surjective $S' \to S$, with $S'|_{S_i}$ lying over S'_i, with $f^{-1}(U)$ dense in S', and with S' normal. The local prolongations glue by Theorem 1.1.

(d) It suffices to verify the theorem after replacing U by a non-empty open subspace $V \subset U$: over integral normal S', a prolongation of $A|_V$ is automatically a prolongation of A.

2 Proof of Gabber's lemma

Let $U \subset S$ and A/U be as in Theorem 1.4, and let d be the relative dimension of A over U. We may and shall assume $d \geq 1$. We seek an integral normal S' and a projective surjective morphism $f : S' \to S$ such that the inverse image of A over $f^{-1}(U)$ extends to a semi-abelian algebraic space over S'. In doing so we may shrink U and replace S by an integral normal scheme S_1 mapping projectively and surjectively to S, replace U by its inverse image, and replace A by its inverse image.

Let $\eta = \operatorname{Spec} K$ be the generic point of U, and let A_η be the fibre of A/U at η. We may assume K infinite, otherwise $S = U = \operatorname{Spec} K$. Choose a projective embedding

$$A_\eta \hookrightarrow \mathbf{P}^N.$$

In the Grassmannian of linear subspaces $H \subset \mathbf{P}^N$ of codimension $d-1$, those for which $H \cap A_\eta$ is a smooth curve form the points of a dense open subset, hence have K-points. This density follows by iteration from the fact that a general hyperplane section is transverse. Choose such an H and put

$$T_\eta = A_\eta \cap H.$$

Over an algebraic closure $\overline{K}$ of K the weak Lefschetz theorem gives

$$H^0(A_{\overline{K}}, \mathbf{Z}_\ell) \longrightarrow H^0(T_{\overline{K}}, \mathbf{Z}_\ell), \qquad H^1(A_{\overline{K}}, \mathbf{Z}_\ell) \hookrightarrow H^1(T_{\overline{K}}, \mathbf{Z}_\ell).$$

Thus $T_{\overline{K}}$ is connected and the morphism

$$\operatorname{Alb}(T_\eta) = \operatorname{Pic}^0(T_\eta) \longrightarrow A_\eta$$

from the Jacobian of T_η to A_η is an epimorphism. In particular T_η has genus $g \geq d$. Temporarily exclude the case $d = 1$; then $g \geq 2$.

After shrinking U there exists a smooth curve T_U of genus $g \geq 2$, embedded in A_U, such that

$$\operatorname{Pic}^0(T_U) \twoheadrightarrow A_U.$$

Applying Lemma 1.5 to the curve T_U over $U \subset S$, and replacing S by the resulting modification (or by the normalization of one irreducible component), we may assume that T_U extends to a family of stable curves T over S.

By Artin's representability theorem, $\operatorname{Pic}^0(T)$ exists as an algebraic space over S. Let

$$P := \operatorname{Pic}^0(T)$$

be the open algebraic subspace in groups parametrizing invertible sheaves on T whose restriction to every irreducible component of every geometric fibre has degree zero. This algebraic space is separated and smooth of finite type over S. Its geometric fibres are extensions of abelian varieties by tori; hence P is a semi-abelian algebraic space over S.

We have therefore obtained, after the allowed modification of S and shrinking of U, a semi-abelian algebraic space P over S such that P_U is abelian and carries an epimorphism

$$u : P_U \twoheadrightarrow A.$$

Let $K_U = \operatorname{Ker}(u)$. It is a subgroup algebraic subspace of P_U. After shrinking U if necessary, we may suppose it flat over U. By flattening by modification, a further modification of S permits us to suppose that the schematic closure K of K_U in P is flat over S; then K is automatically a subgroup algebraic space of P.

For algebraic spaces in groups over a base S, the quotient by a flat closed subgroup algebraic space exists, is compatible with base change, is separated over S, and is flat if the original group space is flat. Let

$$N = P/K.$$

The geometric fibres of N are quotients of extensions of abelian varieties by tori, hence again extensions of abelian varieties by tori. Thus N is semi-abelian, and since $A = P_U/K_U$, it extends A and proves the theorem.

It remains only to handle the case $d = 1$. The same proof applies if one defines a stable curve of genus one as a pointed curve $X \to S$ with section in the smooth locus and with geometric fibres either elliptic curves or nodal cubics. Alternatively, one may define T_η as a hyperplane section of $A_\eta \times A_\eta$.

3 Exorcising stacks

3.1 Recall the definition of a groupoid in schemes. It is a diagram

$$S_2 \substack{\longrightarrow\\[-0.3em]\longrightarrow\\[-0.3em]\longrightarrow} S_1 \rightrightarrows S_0 \tag{3.1.1}$$

with face maps d_i, $i = 0, 1$ and $i = 0, 1, 2$ respectively, such that the square

$$\begin{array}{ccc} S_2 & \longrightarrow & S_1 \\ \downarrow & & \downarrow{\scriptstyle d_1} \\ S_1 & \xrightarrow{d_0} & S_0 \end{array}$$

is cartesian and such that, for every scheme T, the sets $\mathrm{Hom}(T, S_i)$ define a groupoid. The objects are the elements of $\mathrm{Hom}(T, S_0)$, the arrows are those of $\mathrm{Hom}(T, S_1)$, the source and target of an arrow f are $d_1(f)$ and $d_0(f)$, and if the target of f is the source of g, the composition is $g \circ f = d_1\langle g, f\rangle$, where $\langle g, f\rangle \in \mathrm{Hom}(T, S_2)$ is determined by $d_2\langle g, f\rangle = f$ and $d_0\langle g, f\rangle = g$. We denote this groupoid by $S_*(T)$. As T varies, the $S_*(T)$ form a prestack in groupoids.

The quasi-compact separated algebraic stacks are the stacks in groupoids obtained from such a prestack, for the étale topology, when S_0 is quasi-compact, $(d_0, d_1) : S_1 \to S_0 \times S_0$ is finite, and the $d_i : S_1 \to S_0$ are étale.

The quasi-compact separated algebraic spaces correspond to the particular case where $(d_0, d_1) : S_1 \to S_0 \times S_0$ is a closed immersion. Then $S_1 \subset S_0 \times S_0$ is an étale equivalence relation, and the algebraic space is the quotient S_0/S_1.

3.2 Let $g \geq 2$, and say simply stable curve for stable curve of genus g. We explain how to construct a diagram (3.1.1) for the stack $\mathcal{M}$ of stable curves. If T_i is endowed with a stable curve $u_i : X_i \to T_i$ for $i = 1, 2$, write

$$T_1 \times_{\mathcal{M}} T_2 = \mathrm{Isom}(X_1, X_2).$$

This represents the functor sending T to triples (a_1, a_2, φ), where $a_i : T \to T_i$ and $\varphi : a_1^* X_1 \xrightarrow{\sim} a_2^* X_2$. This fibre product is finite over $T_1 \times T_2$.

The Hilbert scheme H of stable curves X endowed with a tricanonical embedding

$$X \hookrightarrow \mathbf{P}^{5g-6}$$

is smooth over $\mathcal{M}$ and maps onto $\mathcal{M}$. The group $\mathrm{PGL}(5g - 5)$ acts on H with finite reduced stabilizers. If S is a disjoint sum of smooth subschemes of H, transverse to the PGL-orbits, of complementary dimension and meeting all of them, then S is étale over $\mathcal{M}$ and maps onto it, hence supplies a diagram (3.1.1). The stack $\mathcal{M}$ is separated of finite type over $\mathrm{Spec}(\mathbf{Z})$. Since deformations of stable curves are unobstructed, it is smooth; this also follows from the smoothness of H over $\mathrm{Spec}(\mathbf{Z})$.

3.3 Let $n \geq 1$. Let $\mathcal{M}^0[n]$ be the algebraic stack classifying proper smooth curves X of genus g over bases S where n is invertible, together with a level-n structure

$$\alpha : \mathrm{Pic}^0(X)_n \xrightarrow{\sim} (\mathbf{Z}/n\mathbf{Z})^{2g}.$$

If $\mathcal{M}^0$ is the open part of $\mathcal{M}$ corresponding to smooth curves, the forgetful morphism

$$\mathcal{M}^0[n] \longrightarrow \mathcal{M}^0[1/n]$$

makes $\mathcal{M}^0[n]$ a finite étale cover of $\mathcal{M}^0[1/n]$.

3.4

Definition 3.1. *Let $\mathcal{M}_n$ be the normalization of $\mathcal{M}$ in $\mathcal{M}^0[n]$.*

In the language of 3.1 this means the following. Choose S_0 étale over $\mathcal{M}$ and mapping onto it, whence a diagram S_*. Over each S_i there is a stable curve X_i, and these form a cartesian system

$$\begin{tikzcd} X_2 \arrow[r, shift left] \arrow[r, shift right] & X_1 \arrow[r, shift left] \arrow[r, shift right] & X_0 \\ S_2 \arrow[r, shift left] \arrow[r, shift right] \arrow[u] & S_1 \arrow[r, shift left] \arrow[r, shift right] \arrow[u] & S_0. \arrow[u] \end{tikzcd}$$

Let S_i^0 be the open subset over which X_i is smooth, and let $S_{i,n}[1/n]$ be the finite étale cover of $S_i^0[1/n]$ given by

$$\mathrm{Isom}(\mathrm{Pic}^0(X_i)_n, (\mathbf{Z}/n\mathbf{Z})^{2g}).$$

Let $S_{i,n}$ be the normalization of S_i in $S_{i,n}[1/n]$. Since normalization commutes with étale base change in this situation, the $S_{i,n}$ again form a diagram (3.1.1), defining $\mathcal{M}_n$. The diagram

$$\begin{tikzcd} S_{2,n} \arrow[r, shift left] \arrow[r, shift right] \arrow[d] & S_{1,n} \arrow[r, shift left] \arrow[r, shift right] \arrow[d] & S_{0,n} \arrow[d] \\ S_2 \arrow[r, shift left] \arrow[r, shift right] & S_1 \arrow[r, shift left] \arrow[r, shift right] & S_0 \end{tikzcd}$$

is cartesian. In particular, for $t \in S_{0,n}(T)$ with image $s \in S_0(T)$ one has an injection of automorphism groups

$$\mathrm{Aut}(t) \hookrightarrow \mathrm{Aut}(s). \tag{3.4.1}$$

3.5

Proposition 3.2. *If $n \geq 3$, then $\mathcal{M}_n[1/n]$ is an algebraic space.*

The essential point is the following lemma.

Lemma 3.3. *Let X be a stable curve over an algebraically closed field k, and let $n \geq 3$ be invertible in k. Every automorphism of X acting trivially on $\mathrm{Pic}^0(X)_n$ is the identity.*

Proof. Write $\mathrm{Pic}^0(X)$ as an extension of an abelian variety by a torus,

$$0 \longrightarrow T \longrightarrow \mathrm{Pic}^0(X) \longrightarrow A \longrightarrow 0,$$

and let L be the character group of T. Every automorphism u of X has finite order. If it acts trivially on $\mathrm{Pic}^0(X)_n$, it acts trivially on A_n and on T_n, hence on L/nL. By Serre's rigidity for A, and by the analogous direct argument for L, it acts trivially on A and on L, hence on T. Thus $u - 1$ sends $\mathrm{Pic}^0(X)$ to T and is therefore zero; u acts trivially on $\mathrm{Pic}^0(X)$, and the conclusion follows from the standard faithfulness result for stable curves. □

Lemma 3.4. *Let S be normal, let $j : U \hookrightarrow S$ be a dense open subset, let A be an abelian algebraic space over U, let N be a semi-abelian prolongation of A over S, and let n be invertible on S.*

(i) The étale sheaf N_n of points killed by multiplication by n injects into $j_* A_n$.

(ii) Every isomorphism over U from A_n to the constant sheaf $(\mathbf{Z}/n\mathbf{Z})^{2g}$ extends to a monomorphism

$$N_n \hookrightarrow (\mathbf{Z}/n\mathbf{Z})^{2g}.$$

Proof. The first assertion follows from the separatedness of N_n over S. Since S is normal, the constant sheaf satisfies

$$(\mathbf{Z}/n\mathbf{Z})^{2g} \simeq j_*(\mathbf{Z}/n\mathbf{Z})^{2g},$$

and the composite

$$N_n \longrightarrow j_* j^* N_n = j_* A_n \longrightarrow j_*(\mathbf{Z}/n\mathbf{Z})^{2g} = (\mathbf{Z}/n\mathbf{Z})^{2g}$$

is the required prolongation. □

Proof of the proposition. Let $T_1 = S_{1,n}[1/n]$. Since T_1 is finite and unramified over $T_0 \times T_0$, it suffices to verify that for every algebraically closed field k, the map

$$T_1(k) \longrightarrow T_0(k) \times T_0(k)$$

is injective. We may suppose n invertible in k, otherwise $T_1(k) = \emptyset$. Because $T_*(k)$ is a groupoid, it suffices to verify that for $t \in T_0(k)$ there is exactly one object of $T_1(k)$ over (t, t), i.e. that $\mathrm{Aut}(t)$ is reduced to the identity.

Let X denote the stable curve obtained by pullback of the universal curve. On $T_i^0 = S_{i,n}[1/n]$ there is by definition a cartesian system of isomorphisms

$$\mathrm{Pic}^0(X_i)_n \xrightarrow{\sim} (\mathbf{Z}/n\mathbf{Z})^{2g}.$$

By Lemma 3.4 it extends over T_i to a monomorphism of étale sheaves. It follows that the morphism from the stack defined by T_* to that defined by S_* factors through the stack of stable curves X endowed with a monomorphism $\mathrm{Pic}^0(X)_n \hookrightarrow (\mathbf{Z}/n\mathbf{Z})^{2g}$. For $t \in T_0(k)$, the injection (3.4.1) has image consisting of automorphisms of the corresponding stable curve acting trivially on $\mathrm{Pic}^0(X)_n$. By Lemma 3.3, such an automorphism is trivial. □

3.6

Corollary 3.5. *If n is divisible by two integers n_i, $i = 1, 2$, prime to one another and each at least 3, then $\mathcal{M}_n$ is an algebraic space.*

It suffices to verify that each $\mathcal{M}_n[1/n_i]$ is an algebraic space, i.e. that the corresponding objects have no non-trivial automorphisms. This follows from (3.4.1) and its factorization

$$\mathcal{M}_n[1/n_i] \longrightarrow \mathcal{M}_{n_i}[1/n_i] \longrightarrow \mathcal{M},$$

together with Proposition 3.5.

3.7 In 3.6, stack language gave an étale atlas defining an algebraic space $\mathcal{M}_n$. The valuative criteria of separatedness and properness, together with the semi-stable reduction theorem, show that this algebraic space is proper over $\mathrm{Spec}(\mathbf{Z})$. One may use this $\mathcal{M}_n$ as the S_0 in the proof of Lemma 1.5. If one wants S_0 to be projective over $\mathrm{Spec}(\mathbf{Z})$, one applies Chow's lemma to $\mathcal{M}_n$. In fact $\mathcal{M}_n$ is projective, by Mumford.

4 Obtaining schemes

In this section we show how to modify and complete the proof of Gabber's lemma so as to obtain schemes rather than algebraic spaces. Another approach was indicated to me by Michel Raynaud: he says he can prove that if X is an S-algebraic space in groups, smooth and separated over S, with connected fibres, then X becomes representable after base change to a suitable blow-up S'/S.

Let $g \geq 2$. For brevity again say stable curve for stable curve of genus g.

4.1 For a smooth curve X/S of genus g, autoduality of the Jacobian

$$\varepsilon : \mathrm{Pic}^0(X) \longrightarrow \mathrm{Pic}^0(X)^*$$

gives a canonical relatively ample invertible sheaf on $\mathrm{Pic}^0(X)$, namely

$$(1, \varepsilon)^*(\text{Poincaré sheaf}).$$

It is symmetric and is trivialized along the zero section. Its formation is compatible with arbitrary base change. Denote it by $\mathcal{L}(X/S)$.

4.2

Lemma 4.1. *It is possible, uniquely up to unique isomorphism, to attach to each stable curve X/S an invertible sheaf $\mathcal{L}(X/S)$ on $\mathrm{Pic}^0(X)$, trivialized along the zero section, compatible with all base change, and equal to the sheaf of 4.1 when X is smooth.*

Proof. It suffices to treat the universal case. With the notation of 3.4, choose X_*/S_*. We seek invertible sheaves $\mathcal{L}(X_i/S_i)$ on the $\mathrm{Pic}^0(X_i)$, trivialized along the zero section, extending the sheaves of 4.1 over the open subset where X_i is smooth and forming a cartesian system under the maps $S_i \to S_j$. Each S_i is smooth over $\mathrm{Spec}(\mathbf{Z})$, hence regular, and so is $\mathrm{Pic}^0(X_i)$, which is smooth over S_i. An extension exists by taking the determinant of any coherent extension. Correcting by the inverse image of its restriction to the zero section gives a trivialized extension. Since $\mathrm{Pic}^0(X_i)$ has geometrically irreducible fibres, such an extension is unique up to unique isomorphism, and these unique isomorphisms give the required compatibility. $\square$

4.3

Proposition 4.2. *For a stable curve X/S, the invertible sheaf $\mathcal{L}(X/S)$ of 4.2 is relatively ample. In particular,* $\mathrm{Pic}^0(X)$ *is quasi-projective over S and is a scheme if S is a scheme.*

As before it suffices to verify this in a universal case; one may suppose S regular and the open subset $S^0 \subset S$ over which X is smooth dense. If one already knew that $\mathrm{Pic}^0(X)$ were a scheme, the assertion would follow from Raynaud's theorem. Unfortunately that proof uses the scheme hypothesis essentially in order to construct, locally on S, relatively ample divisors as complements of affine open subsets.

4.4

Lemma 4.3. *Let X be a geometrically reduced connected curve over a separably closed field k. There exists a divisor D, contained in the smooth locus of X, of degree*

$$g = \dim H^1(X, \mathcal{O}_X),$$

such that

$$H^0(X, \mathcal{O}_X(D)) = k, \qquad H^1(X, \mathcal{O}_X(D)) = 0.$$

Proof. The assertion on H^0 follows from the other assertion by Riemann-Roch. If ω is the dualizing sheaf, then $\dim H^0(X,\omega) = g$, and by duality $H^1(X,\mathcal{O}_X(D)) = 0$ is equivalent to $H^0(X,\omega(-D)) = 0$. The sheaf ω has no embedded component. We may therefore construct by induction divisors D_i of degree i, contained in the smooth locus, with

$$\dim H^0(X,\omega(-D_i)) = g - i,$$

by taking $D_0 = 0$ and $D_{i+1} = D_i + P_i$, where P_i is a smooth k-point at which not all sections of $H^0(X,\omega(-D_i))$ vanish. Take $D = D_g$. □

4.5

Lemma 4.4. *Let $f : X \to S$ be projective, with geometrically reduced connected fibres. Locally for the étale topology on S, there is an open subset*

$$V \subset \mathrm{Pic}^0(X)$$

meeting every fibre and which is a scheme.

Proof. One reduces to the case where S is strictly local with closed point s, and it suffices to construct an open $V \subset \mathrm{Pic}^0(X)$ which is a scheme and meets the fibre over s. By Lemma 4.3, choose a divisor D_s of degree $g = \dim H^1(X_s,\mathcal{O}_{X_s})$ on X_s, contained in the smooth locus, and satisfying $H^1(X_s,\mathcal{O}(D_s)) = 0$. Extend it to a relative Cartier divisor D on X, possible because S is local henselian. Then

$$f_*\mathcal{O}(D) = \mathcal{O}_S, \qquad R^1 f_*\mathcal{O}(D) = 0.$$

Let V' be the open subset of $\mathrm{Pic}^0(X)$ corresponding to invertible sheaves $\mathcal{F}$ such that $R^1 f_*\mathcal{F}(D) = 0$. For such $\mathcal{F}$, $f_*\mathcal{F}(D)$ is free of rank one and commutes with all base change. Let $V \subset V'$ be the open corresponding to those $\mathcal{F}$ for which the zero locus of a basis section of $f_*\mathcal{F}(D)$ is a relative Cartier divisor $D(\mathcal{F})$ contained in the smooth locus X^0. By construction V contains the zero section. The map

$$\mathcal{F} \longmapsto D(\mathcal{F})$$

identifies V with an open subset of $\mathrm{Sym}^g(X^0)$, and hence V is a scheme. □

4.6

Lemma 4.5. *Under the hypotheses of Lemma 4.4,* $\mathrm{Pic}^0(X)$ *is locally a scheme for the étale topology on S.*

Proof. Again suppose S strictly local. Put $P = \mathrm{Pic}^0(X)$. It is a smooth algebraic space over S, pure of relative dimension g. Let V be as in Lemma 4.4. It suffices to show that P is the union of the translates gV for $g \in P(S)$. Reducing to the case where S is essentially of finite type over $\mathbf{Z}$, one may count dimensions. Let P^n be the n-fold fibre product of P over S. For a geometric point t of S and a geometric point $\gamma = (\gamma_1,\ldots,\gamma_n)$ of P^n over t, say that γ is allowed if the translates $\gamma_i V$ cover P_t. The allowed points form a constructible subset Y of P^n. If Z is its complement, then fibre by fibre Z has codimension at least $n - g$. For $n - g > \dim S$, the closure of Z cannot contain the whole special fibre, so one finds a section γ of $P^n - \overline{Z}$, and then P is the union of the $\gamma_i V$. □

4.7 In Lemmas 4.4 and 4.5, the projectivity hypothesis on f may be replaced by properness: locally on S for the étale topology, f is automatically projective.

4.8

Proof of Proposition 4.2. The problem is local on S for the étale topology. By Lemma 4.5, one can apply Raynaud's theorem giving relative ampleness in the scheme case. $\square$

4.9 The proof of Raynaud's theorem seems to use the scheme hypothesis only through the conclusion of Lemma 4.4. This would allow one to avoid Lemma 4.5.

4.10

Proposition 4.6. *Gabber's lemma remains true with "semi-abelian algebraic space" in the conclusion replaced by "semi-abelian scheme."*

Proof. Proceeding as in Section 2, one reduces to the case where there is a stable curve X over S, smooth over U, and an epimorphism over U

$$u : \mathrm{Pic}^0(X) \longrightarrow A$$

whose kernel K_U extends to a flat subgroup algebraic space K of $\mathrm{Pic}^0(X)$. It remains to show that

$$P := \mathrm{Pic}^0(X)/K$$

is a scheme.

Over U, Poincaré complete reducibility gives a morphism

$$v : A \longrightarrow \mathrm{Pic}^0(X)$$

such that uv is multiplication by an integer $n \geq 1$ on A. By the extension result of 1.3, v extends to a morphism, again denoted v, from P to $\mathrm{Pic}^0(X)$. By density of U in S, one still has $uv = n$. Since multiplication by n on P is quasi-finite, v is quasi-finite, hence quasi-affine. Since $\mathrm{Pic}^0(X)$ is a scheme, P is a scheme. $\square$

I thank M. Raynaud and O. Gabber for a critical reading of a first version of this text.

Note

(1).1 Let $S = \mathrm{Spec}\, A$ be affine, and let $U \subset S$ be an open subscheme which is a finite union of principal opens $U_i = \mathrm{Spec}\, A[f_i^{-1}]$. If F is a closed subscheme of U, defined in U_i by an ideal $I_i \subset A[f_i^{-1}]$, then the schematic closure $\overline{F}$ of F in S is the closed subscheme defined by

$$I = \bigcap_i \rho_i^{-1}(I_i),$$

where $\rho_i : A \to A[f_i^{-1}]$ is the canonical map. If $S' = \mathrm{Spec}\, A'$ is flat over S, then the inverse image of $\overline{F}$ in S' is again the schematic closure of the inverse image of F. Hence schematic closure commutes with étale localization and so makes sense for algebraic spaces.

More generally, for $f : X \to S$ quasi-compact, the closed schematic image $f(X)^-$ is the closed subscheme defined by the ideal of elements $a \in A$ whose inverse image on X is zero. For f the inclusion of F as above, this recovers schematic closure. The same base-change compatibility holds for schematic images.

(1).2 Let S be a noetherian algebraic space, and let X and Y be two algebraic spaces over S, each endowed with a section e. Let $V \subset X$ be open, put $U = e^{-1}(V) \subset S$, and let

$$f_V : V \longrightarrow Y$$

be an S-morphism sending e to e. For each n, f_V induces a morphism from the n-th infinitesimal neighbourhood of e in V to the corresponding neighbourhood in Y_U. Passing to the inverse limit gives a morphism

$$f_V^\wedge : X_U^\wedge \longrightarrow Y_U^\wedge$$

between formal completions along the sections.

Proposition 4.7. *Assume X smooth over S and Y separated of finite type.*

(i) If U contains all points of S of depth at most 1, then $f_V^\wedge$ extends uniquely to a morphism $f^\wedge : X^\wedge \to Y^\wedge$ between the formal completions of X and Y along the sections.

(ii) Let Γ be the schematic closure in $X \times_S Y$ of the graph $\Gamma_V \subset V \times_S Y$ of f_V. If $f_V^\wedge$ extends to $f^\wedge : X^\wedge \to Y^\wedge$ and U contains all points of S of depth 0, then Γ is étale over X along the section e, and its formal completion there is the graph of $f^\wedge$.

The proof is local on S for the étale topology; thus one may suppose S affine. Then the infinitesimal neighbourhoods of the section in X and Y are affine schemes, and (i) is proved as in Raynaud. Smoothness of X/S ensures that the infinitesimal neighbourhoods of e in X are finite and flat over S.

We use the following lemma for (ii).

Lemma 4.8. *Let $S = \operatorname{Spec} A$ with A a complete noetherian local ring, let*

$$X = \operatorname{Spec} R[[T_1, \ldots, T_n]],$$

let e be the zero section, let V be an open subset of X such that $U = e^{-1}(V)$ contains all associated points of S, and let $F \subset V$ be a closed subscheme containing all infinitesimal neighbourhoods of $e(U)$ in V. Then $F = V$.

Proof. It suffices to show that X is the schematic closure of F in X. If

$$f = \sum a_I T^I$$

vanishes on F, then each coefficient a_I vanishes on $U \subset S$, hence is zero, and $f = 0$. □

Proof of Proposition 4.7(ii). Let s be a point of S. Write $X_s^\wedge$ for the completion of X at $e(s)$, and let $V_s^\wedge$ and U_s be the inverse images of V and U. The morphism $f^\wedge$ induces $X_s^\wedge \to Y$, and f_V induces $V_s^\wedge \to Y$. These two morphisms coincide on a closed subscheme containing all infinitesimal neighbourhoods of $e(U_s)$, hence coincide by Lemma 4.8. The inverse image of Γ_V in $X_s^\wedge \times_S Y$ is therefore the graph of $f^\wedge|_{V_s^\wedge}$, whose schematic closure is the graph of $f^\wedge : X_s^\wedge \to Y$. Since schematic closure commutes with the flat base change $X_s^\wedge \times_S Y \to X \times_S Y$, it follows that after the base change $X_s^\wedge \to X$, the space Γ is étale over X. Hence Γ is étale over X at $e(s)$. □

(1).3

Corollary 4.9. *Let G and H be algebraic spaces in groups over S, let U be an open subspace of S, and let*

$$f_U : G_U \longrightarrow H_U$$

be a morphism. Assume that G is smooth over S with connected fibres, that H is of finite type and separated over S, and that U contains all points of S of depth 0. If the formal completion of f_U along the zero sections extends to a morphism of formal completions

$$G^\wedge \longrightarrow H^\wedge,$$

then f_U extends uniquely to a morphism $G \to H$.

If U contains all points of depth at most 1, the existence of the formal extension is guaranteed by Proposition 4.7(i).

Proof. Uniqueness follows from the fact that U is schematically dense in S, hence G_U is schematically dense in G. For existence, let Γ_U be the graph of f_U, let $\Gamma \subset G \times_S H$ be its schematic closure, and let Γ^0 be the open subset of Γ where Γ is étale over G. By Proposition 4.7(ii), Γ^0 contains the zero section of $G \times_S H$. For every section γ of Γ, left translation by the element of $G_U \times_S H_U$ determined by $\gamma|_U$ sends Γ_U to itself. By transport of structure, left translation by γ sends Γ^0 to itself. After the flat base change Γ^0/S, this shows that Γ^0 is a subgroup algebraic space of $G \times_S H$.

The image of Γ^0 in G is an open subgroup algebraic space, hence is all of G because G/S has connected fibres. Since Γ^0/G is étale and separated, the function assigning to x the number of geometric points in the fibre over x is lower semicontinuous. It is at least one everywhere and equals one over G_U, hence is constantly one. Therefore $\Gamma^0 \to G$ is étale, surjective, and radicial, hence an isomorphism. The required extension is the composite

$$G \xleftarrow{\sim} \Gamma^0 \longrightarrow H.$$

□

(1).4 The preceding proposition should remain true if the hypothesis that G/S is smooth is weakened to flatness of finite type, and if the hypothesis that H is separated over S is omitted.

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Sigla

EGA denotes *Éléments de géométrie algébrique*, by A. Grothendieck and J. Dieudonné. SGA denotes *Séminaire de géométrie algébrique du Bois-Marie*.